

教學實作：擴充 DHT11 溫濕度感測器功能

步驟 1：修改硬體定義提示詞 (Devices Definition)

讓 AI 認識 DHT11 感測器及其物理特性。

AMB82-mini ： PIN 8

HUB 8735 Ultra ： PIN 20

DHT11 Temperature & Humidity Sensor

- Pin mapping: depends on development board
 - AMB82-mini: PIN 8
 - HUB 8735 Ultra: PIN 20
- Measures: temperature (°C) and relative humidity (%)
- Read mode: single trigger, returns two integer values
- Temperature range: 0–50 °C
- Humidity range: 20–90 % RH
- Physical Rules: Values are integers. Sensor requires ~1 s between reads.

步驟 2：定義工具 JSON 規範 (Tools Definition)

在 toolsDefinition 中加入 /dht11 指令的格式規範。

Reading the DHT11 temperature and humidity sensor:

```
{
  "type": "tool_call",
  "method": "/dht11",
  "params": {
  }
}
```

Success response:

```
{
  "status": "success",
  "method": "/dht11",
  "temperature": <temperature value>,
  "humidity": <humidity value>,
  "workId": "<system-provided>"
}
```

Error response:

```
{
  "status": "error",
  "method": "/dht11",
  "reason": "<error reason>",
  "workId": "<system-provided>"
}
```

步驟 3：底層 C++ 程式碼實作

在專案中新增 `tool_dht11` 驅動函數。

```
#include "DHT.h"
#define DHTPIN 20
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);

String tool_dht11(String workId) {
    float h = dht.readHumidity();
    float t = dht.readTemperature();
    if (isnan(h) || isnan(t)) {
        return "{\"status\":\"error\","
            "\"method\":\"/dht11\","
            "\"reason\":\"dht11_read_failed\","
            "\"workId\":\"\" + workId + "\"}";
    }

    return
        "{\"status\":\"success\","
        "\"method\":\"/dht11\","
        "\"temperature\":\"" + String(t) + "\","
        "\"humidity\":\"" + String(h) + "\","
        "\"workId\":\"\" + workId + "\"}";
}

void setup() {
    // ... 原有初始化內容 ...
    delay(1000); // 等待感測器穩定

    dht.begin();
}
```

步驟 4：更新工具分發器邏輯 (executeTool)

將 /dht11 方法註冊進系統的分發流程。

```
void executeTool(String workId, String command, JsonObject params, bool reCheck = true) {

    if (command == "/digitalwrite" || command == "/analogwrite") {
        // ... 原有內容不變 ...

    }

    else if (command == "/dht11") {

        String response = tool_dht11(workId);

        if (xSemaphoreTake(stateMutex, MUTEX_TIMEOUT_TICKS) == pdTRUE) {
            historicalMessages += buildGeminiMessage("user", command + timestamps);
            historicalMessages += buildGeminiMessage("model", response + timestamps);
            executeToolHistory += workId + " " + command + " [ " + response + " ]\n";
            xSemaphoreGive(stateMutex);
        }

        evaluateWorkflowContinuation(workId, reCheck);

    }
    else if (command == "/digitalread" || command == "/analogread") {
        // ... 原有內容不變 ...
    }
    // ... 其餘 tools 不變 ...
}
```